



R.B. Brown

Dalhousie University

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Department of Psychology

**26th Annual
In-House Convention**

27-28 April 2000

Time	Thursday	Friday
8:30	Introductory remarks Richard Brown	Marsha Penner, I. Valdivia ...
8:45	Cathy Forestell	Elliott Marchant
9:00	Sharon Clark	Marleen de Groot
9:15	Anna Lee	John Barresi
9:30	Scott Fairless	Christensen et al.
9:45	BREAK	BREAK
10:00	Ed Johnson	Jennifer Klages
10:15	Amelia Hunt and Ray Klein	Mike Ellery
10:30	Jason Ivanoff and Amelia Hunt	Kelly Shoemaker
10:45	Ray Klein and Bruce Dick	Robin Patterson
11:00	Jeff Hancock and Ray Klein	Melanie McFadyen
11:15	BREAK	BREAK
11:30	Mike Woodworth	Lianne Stanford
11:45	Paddy McMullen	Heather Schellinck
12:00	Jana Padhorna	William Brown
12:15	Peggy Ruyak	Ron Hoffman and RE Brown
12:30	LUNCH	LUNCH
12:45		
1:00		
1:15		
1:30		
1:45	Jeanette McGlone	Ron Hoffman, V Hannon ...
2:00	Bev Butler	Ray Klein, David Shore, J Gati
2:15	Sandy Sparkes	David Shore
2:30	Connolly Symposium	Joseph MacInnes
2:45		Tracy Taylor-Helmick
3:00		Susan Boehnke
3:15		Ivan Valdivia
3:30	BREAK	BREAK
3:45	Connolly Symposium	Symposium on Consciousness (introduction by Ray Klein)
4:00		
4:15		
4:30		
4:45		
	Thursday posters	Friday Posters
	Tamara Bond	Elizabeth McLaughlin
	Gail Eskes	Nicole Blanchet
	Angie Birt	Karen Jaskie
	Mauro Vieira	Sean McCabe
	Mary-Ellen Large	Shana Nichols
	Jen Stapleton	Marsha Penner

General notes of Japan diseases

Thursday, April 27, 2000 (LSC 4258)

8:30 **Introduction to the In-House Convention** —Richard Brown

8:45— 9:45 **Session 1.** Chair: Sarah Giles

- T1) Cathy Forestell
- T2) Sharon Clark
- T3) Anna Lee
- T4) Scott Fairless

9:45—10:00 **Break**

10:00—11:15 **Session 2.** Chair: Jennifer Stapleton

- T5) Ed Johnson
- T6) Amelia Hunt
- T7) Jason Ivanoff
- T8) Ray Klein and Bruce Dick
- T9) Jeff Hancock

11:15—11:30 **Break**

11:30—12:30 **Session 3.** Chair: Shawnee Basden

- T10) Mike Woodworth
- T11) Paddy McMullen
- T12) Jana Podhorna
- T13) Peggy Ruyak

12:30— 1:45 **Lunch Break and Posters**

- P1) Tamara Bond
- P2) Gail Eskes
- P3) Angie Birt
- P4) Mauro Vieira
- P5) Mary-Ellen Large
- P6) Jennifer Stapleton

1:45— 2:30 **Session 4.** Chair: Sean McCabe

- T14) Jeannette McGlone
- T15) Beverly Butler
- T16) Sandy Sparkes

2:30— 3:30 **Symposium--The Wonderful World of Cognitive Neuroscience: Basic and Clinical**

- T17) Randy Newman
- T18) Ryan D'Arcy
- T19) Jingtian Wang
- T20) Frank MacMaster

3:30— 3:45 **Break**

3:45— 5:00 **Symposium continued...**

- T21) Shakeela Khan
- T22) Deanna Braaksma
- T23) Michael Houlihan
- T24) Yannick Marchand
- T25) Catherine Mayo

Friday, April 28, 2000 (LSC 4258)

8:30— 9:45 **Session 6.** Chair: Sarah Mathieson

- T26) Marsha Penner
- T27) Elliott Marchant
- T28) Marleen deGroot
- T29) John Barresi
- T30) C. Christensen

9:45—10:00 **Break**

10:00—11:15 **Session 7.** Chair: Christianne Hawken

- T31) Jennifer Klages
- T32) Mike Ellery
- T33) Kelly Shoemaker
- T34) Robin Patterson
- T35) Melanie McFadyen

11:15—11:30 **Break**

11:30—12:30 **Session 8.** Chair: Melanie McFadyen

- T36) Lianne Stanford
- T37) Heather Schellinck
- T38) Will Brown
- T39) Ron Hoffman

12:30—1:45 **Lunch Break and Posters**

- P7) Elizabeth McLaughlin
- P8) Nicole blanchet
- P9) Karen Jaskie
- P10) Shana Nichols
- P11) Marsha Penner
- P12) Sean McCabe
- P13) Randy Newman

1:45— 3:30 **Session 9.** Chair: Tarek Loubani

- T40) Ron Hoffman
- T41) Ray Klein
- T42) David Shore
- T43) Joe MacInnes
- T44) Tracy Taylor-Helmick
- T45) Susan Boehnke
- T46) Ivan Valdivia

3:30— 3:45 **Break**

3:45— 5:00 **Symposium—Consciousness**

- T47) Ray Klein
- T48) John Barresi
- T49) Deanna Braaksma
- T50) John Christie
- T51) Bill Matheson

Welcome to the 2000 Psychology and Neuroscience In-House Convention

Welcome to the Twenty-Sixth Annual Dalhousie University Department of Psychology In-House Convention. This year's meeting is being hosted by the students and Faculty of our Graduate Program in Psychology. The convention planning committee consists of the following individuals: N. Blanchet, M. Ellery, K. Jaskie, R. Newman, K. Shoemaker, J. Wang, and M. Yoon. A special thanks the Suzanne King for efforts above and beyond the call of duty, to David Rioux for his assistance with the In-House Convention web page, and to Susan Boehnke for her assistance with the audio-visual equipment.

Friendly Reminders

- Please bring your own mug for coffee or tea!
- All presentations are listed in the name of the first author from the abstracts submitted.
- There will be a social at the end of Thursday and pizza and beer at the end of Friday, in the 2nd Floor Lounge.

An Historical Introduction to the In-House Conference

Richard Brown

This presentation will use photographs from the past 25 years to introduce the In-House. Photos of past faculty, students, and PDF's in their most embarrassing moments will be presented.

T1. Is beauty in the eye of the beholder? Restraint, body shape and weight in attractiveness ratings of self and others.

Catherine A. Forestell, Tara M. Humphrey and Sherry H. Stewart

The present study tested the degree to which body-shape characteristics of line drawings influence restrained and nonrestrained eaters' ratings of attractiveness for female figures. Female line drawings (Tassinari & Hensen, 1998, *Psychological Science*: 9, 150-155), which varied systematically in weight and waist and hip circumference, were used as stimuli. Thirty-two female undergraduate Psychology students (15 nonrestrained and 17 restrained, according to the Restrained Eating Scale; Herman & Polivy, 1980, *Obesity*, pp 208-225), were asked to rate these stimuli in attractiveness, identify the ideal body figure, and the figure which best represented their own body type. Restrained eaters did not differ from nonrestrained eaters in their ratings of attractiveness nor in their perception of the ideal female figure. However, after controlling for body weight and shape, it was found that restrained eaters did perceive themselves as larger than nonrestrained eaters. Thus it appears that the motivation to diet in restrained eaters may stem from an overestimation of their perceived body size.

T2. Form or Function? A body-image assessment tool

Sharon Clark, Joanne Gusella and Erica van Roosmalen

A negative body image is one of the primary features of an eating disorder and often develops from an exclusive focus on body appearance, or the form of one's body, rather than considering the functional aspects of how the body works. We developed and conducted testing of a new assessment measure of body image. This measure compares two dimensions of body image perceptions: form and function. It was hypothesized that if the functional activities of the body are emphasized, and contrasted to what is liked about the body, the discrepancy between form and function would become apparent. This may be an important component in changing cognitive distortions involved in body image problems while beginning to create a sense of appreciation for what one's body does rather than what it looks like. Results of pilot testing with the measure will be presented.

head + hair = 2 apcs?

T3. Effects of lesions of the medial preoptic area on parental behaviour in male and female California mice, *Peromyscus californicus*

Anna W. Lee & Richard E. Brown

Peromyscus californicus are monogamous and biparental mice. The male shows all aspects of parental care, except lactation, as soon as the pups are born. Thus, they are an ideal species in which to compare the neuroanatomical basis of maternal and paternal behaviour. The present study investigated the effects of lesions in the medial preoptic area on the expression of parental behaviour in male and female California mice. Electrolytic or sham lesions were performed two days postpartum on a male or female from each male-female pair. Mice were tested for parental responsiveness for 10 days, beginning one day post surgery. Preliminary results suggest that medial preoptic lesions disrupt parental behaviour in both male and female mice.

T4. Odour-induced *egr-1* expression in the rat olfactory bulb

Daemon Scott Fairless

The expression of immediate early genes (IEGs) have allowed for the mapping of neuronal activation. Both *c-fos* and *egr-1* have been widely used to this end. While expression of *c-fos* seems to be induced as a function of neural activation, there is some evidence that *egr-1* expression is induced only by novel stimuli. Songbirds in captivity show an increase in *egr-1* expression when presented with a novel conspecific song. Induction of *egr-1* decreases with repeated presentations of the same song but can be re-elicited with the presentation of a novel conspecific song.

In this pilot study the effects of a novel odour on the expression of *egr-1* in the rat olfactory bulb are analyzed. Animals were exposed to a novel odour (1% propionic acid) for an hour and a half and then sacrificed. Controls were sacrificed without any novel odour stimulation. Immunoreactivity against *egr-1* in the main olfactory bulbs of each group is compared.

Mouse vs Rat.

T5. Adult egocentrism: Conceptualization and a new measure

Ed Johnson, Department of Psychology, University of Manitoba

Egocentrism is the tendency to construe events and persons according to private knowledge, without appreciating the privileged character of such construal. Although pervasive in young children, it is proposed that habitual egocentrism in adults reflects deficient development and awareness of the self-system. Accordingly, it is hypothesized that dispositional egocentrism in adults should be associated with inadequate self-knowledge, low self-regard, and heightened self-preoccupation, along with a reduced capacity for understanding and empathizing with others. Two studies of college students are presented in support of this hypothesis. In Study 1, a self-report measure of the disposition to respond egocentrically, the Egocentric Response Scale (ERS) is introduced. Greater egocentrism on the ERS was associated with alexithymia, lower self-esteem, rumination about the self, public self-consciousness, and immature empathy. Study 2 showed that the ERS correlated positively with a measure of attitudinal egocentrism, and that on observer versions of both scales observers' ratings exhibited consensus about targets' egocentrism, and correlated positively with self-ratings of egocentrism. Both scales correlated negatively with measures of self-knowledge, mature empathy, and decoding others' emotions from nonverbal signals.

T6. Eliminating the costs of task set reconfiguration

Amelia Hunt and Raymond Klein

Given insufficient time to fully prepare for a switch in task, a deterioration in performance on the first trial of a new task would be expected. The interest of researchers has been captured by the residual switch costs that surprisingly remain despite sufficient time to prepare. We used a very simple task to investigate the costs to reaction time and accuracy associated with changing between two different instructional sets every eight trials. Subjects responded to left and right visual targets by making either spatially compatible or incompatible button press responses. Subjects were cued which of these responses to make by the color of a border appearing on the perimeter of the display. The large costs to reaction time found at a 200 ms border-target SOA were completely eliminated at longer intervals between the border and the stimulus, but costs to accuracy remained. An identical task using saccades towards or away from the target in the place of button presses resulted in the elimination of costs to both accuracy and reaction time. This finding does not support the prevailing hypothesis that a portion of the task set reconfiguration process must await the arrival of the imperative stimulus to be completed. The differences between the saccadic and manual task switching results are discussed with respect to the compatibility of these otherwise identical tasks.

T7. Switch costs without task switching?

Jason Ivanoff & Amelia Hunt

A change in task initially results in detriments to performance. These "switch costs" have been commonly attributed to the re-configuration of task set by way of executive control processes. However, it is not known whether a switch in task, per se, is responsible for switch costs. In this investigation, task instruction did not change and accordingly no switch costs should be observed if they are only due to a re-configuration of task set. Responses were based entirely on the non-spatial identity of targets, and spatial location was completely irrelevant to the correct performance of the task. Series of spatially corresponding and non-corresponding stimulus-response ensembles alternated within an experiment and the task set remained stable over the duration of the experiment. Although there were no changes in task, we observed robust switch costs at two relatively long response-stimulus intervals. The results call into question the general idea that switch costs must be the result of a re-configuration of task set.

T8. Using rapid serial visual presentation (RSVP) to explore the temporal dynamics of reflexive attention shifts.

Ray Klein and Bruce Dick

The immediate effect of a cue in the periphery (e.g., a luminance increment) is the facilitation of processing in the vicinity of the cue. Although it is generally agreed that this facilitative effect reflects the consequence of an exogenously generated shift of attention to the cued location, some investigators have claimed that the facilitation reflects a response bias rather than a genuine improvement. When the cue is uninformative and there is, therefore, no reason to keep attention at the cued location, the facilitation is followed by a long lasting inhibitory aftereffect at the cued location which has been called inhibition of return (IOR). There is considerable interest in determining whether IOR's primary effect is to delay perceptual processing, responding, or shifting of attention. The RSVP paradigm, originally developed in the 1970's and recently popularized in studies of the "attentional blink," is ideally suited to contribute to this debate because of its emphasis on identification accuracy (minimal demands are placed on the motor system) and its temporal precision. The temporal dynamics of the facilitation and inhibition that follow an uninformative peripheral cue, as obtained from the dual-RSVP-stream peripheral cuing paradigm that we have developed for this purpose, will be illustrated.

T9. Made you look - obligatory processing of faces

Jeff Hancock & Ray Klein

Considerable converging evidence from a wide range of fields (neurophysiology, neuropsychology, cognitive and developmental psychology) suggests that faces represent a special class of visual patterns handled by a system separate from other visual processing systems. The present study attempts to extend research on face perception by examining whether faces presented in the visual field capture attention in a stimulus-driven fashion. Black and white high contrast face and non-face (inverted or scrambled faces) images were presented in a dual stream Rapid Serial Visual Presentation (RSVP) paradigm. The streams consisted of randomly selected letters with a single target digit placed pseudo-randomly in one of the streams. Overall, when the digit was presented around the same time as the face (or shortly thereafter) its identification was better when the face was adjacent to the stream with the digit than when the face was near the opposite stream. The opposite pattern emerged as the interval between the appearance of the face and the digit increased. This pattern corresponds roughly with the prototypical pattern of Inhibition of Return and suggests that faces may capture attention. However, significant laterality effects were observed suggesting that the role of attention in perceiving faces may be differential across visual hemifields. These laterality effects are discussed in terms of recent neuroimaging and neuropsychological data suggesting that the perception of faces is highly lateralized.

T10. Truth, Lies, and Videotape: An Investigation of the Ability of Federal Parole Officers and University Students to Detect Deception

Mike Woodworth & Steve Porter, Angela R. Birt

The ability of a group of Canadian federal parole officers to detect deception was investigated over the course of two days of lie detection training. To isolate factors relating to detection accuracy, three groups of undergraduate participants made judgments on the same 24 videotapes: (1) a feedback group which received feedback on accuracy following each judgment; (2) a feedback + cue information group which was given feedback and information on empirically-based cues to deception; and (3) a control group which did not receive feedback or cue information. Results indicated that at baseline all groups performed at or below chance levels. However, overall, all experimental groups (including the parole officers) became significantly better at detecting deception than the control group. The results indicate that detecting deceit is difficult but training and feedback can enhance detection skills.

T11. Individual differences in the development of face recognition: Support for a maturational change at age eight.

Paddy McMullen, Phil Dunham and Fran Dunham

Carey & Diamond (1977) suggested that before the age of 8, children recognize a face by its individual features. After this age, a whole-face (holistic) strategy has developed. Boys and girls aged 7 and 11 performed tests of Mooney faces (a test of holistic processing), Embedded figures (a test of feature-based processing) and upright and inverted face recognition. Children aged 7, successfully recognized upright faces but were at chance at the recognition of inverted faces. More importantly, upright face recognition correlated with Embedded figures performance ($r = 0.56, p < 0.01$) and not with Mooney face performance. Children aged 11, recognized upright and inverted faces above chance levels. Notably, their results showed the opposite pattern to those of the younger children. Upright face recognition now correlated with Mooney face performance ($r = 0.45, p < 0.05$) and not with Embedded figures performance. The maturational hypothesis of face recognition was tested from an individual differences approach and supported.

T12. The Effects of a Novel Psychoactive Drug Flibanserin on Ultrasonic Vocalizations in Rat Pups

Jana Podhorna and Richard Brown

This study compared the effects of flibanserin, a novel 5-HT_{1A} agonist/5-HT_{2A} antagonist; diazepam, a traditional anxiolytic; and imipramine, a traditional antidepressant, on separation-induced ultrasonic vocalizations (USVs), locomotor behavior, negative geotaxis and body temperature of 7-8-day-old rat pups. Flibanserin (5, 10, 25 and 50 mg kg⁻¹) reduced USVs but had no effects on locomotor behavior or geotaxic response. Low doses of flibanserin (0.5, 1, 2 and 4 mg kg⁻¹) had no effect on any behavior. Diazepam (0.25, 0.5, 1 and 2 mg kg⁻¹) not only reduced the USVs but also increased rolling and increased the latency of the geotaxic response. Imipramine (10, 15, 20 and 30 mg kg⁻¹) reduced USVs, increased total locomotor activity and rolling but had no effect on negative geotaxis. None of the drugs altered body temperature. Our data showed that flibanserin is as effective in reducing the USVs as diazepam and imipramine but has a lower incidence of motor impairment side effects. This suggests that flibanserin might be effective for the treatment of mood disturbances such as anxiety.

T13. Recovery Sleep Patterns Following Treatment of Patients with Obstructive Sleep Apnea

Peggy Ruyak, Ben Rusak and Debbie Morrison

The primary purpose of this investigation was to study the effects of continuous positive airway pressure (CPAP) on the patterns of REM sleep and Slow Wave Sleep (SWS) in patients with obstructive sleep apnea (OSA). Analyses of baseline and CPAP treatment data from 38 patients with OSA included sleep stage, CPAP pressure, and number of arousals during the treatment night. The study involved the re-analysis of existing data from patients treated with CPAP on a second night following an initial baseline night of polysomnographic recording. Data were scored to allow for examination of the detailed time course of occurrence of NREM and REM recovery sleep during the treatment night. The results of the experiment will further describe the phenomenon of REM and SWS rebound during the first night of CPAP treatment.

T14. Neurobehavioural outcomes in men and women undergoing coronary artery bypass surgery with or without cardiopulmonary bypass

Jeannette McGlone, Gregory Hirsch, Jean-Francois Legare, and Karen Buth

Replacing the clogged arteries feeding the heart is a well-established operation for patients with ischemic heart disease. It is called coronary artery bypass grafting (CABG). The traditional methods of placing the patient on a heart-lung machine during the operation will be compared with a newer technique of operating on the beating heart. This talk reviews the literature and outlines a randomized, prospective, clinical trial comparing the neuropsychological and quality of life outcomes of patients undergoing these two techniques. The aim of this study is to evaluate longer term (six month) psychosocial outcomes. For the first time, the goal is to investigate the outcomes of women compared to men. Students looking for comprehensive projects over the next 3 years, persons interested in critiquing the proposal, and the mildly curious are most welcome.

T15. Effects of passive limb movement on the Visual Scanning abilities of Neglect subjects in Two Coordinate Spaces

Beverly Butler^{1,2}, Gail Eskes^{1,2}, E. Harrison^{1,2}, & Alison MacDonald², Dalhousie University¹ and Queen Elizabeth II Health Sciences Centre²

Hemispatial neglect is a deficit in directing attention/movement toward the contralesional space. Robertson (1995) reported that passive left limb movement did not improve visual scanning in a single case of neglect, and suggested that active limb movement was required. Contrary to this finding, Ladavas et al. (1997) and our own pilot study, showed that passive movement may improve visual scanning in neglect patients in peripersonal space. The present study attempted to replicate and extend these passive movement effects. Neglect subjects (n=6), right-hemisphere damaged non-neglect controls (n=6) and non-brain-damaged controls (n=5) detected visual targets in peripersonal and extrapersonal space under conditions of no movement and passive left limb movement (produced by functional electrical stimulation) in left hemispace. We were unable to replicate the passive movement effects on visual scanning in this group of neglect patients in peripersonal or extrapersonal space. However, the neglect group showed a gradient of decreasing target detection from right to left, which was not evident in either of the control groups ($p < 0.05$).

T16. The Influences of Child Age and Parent Informant on the Associations Between Parenting Practices and Child Behaviour Problems

Sandie Sparkes and Ann Wasekush.

This project investigated the influences of child age and parent informant (mother versus father report) on the associations between parenting practices and conduct problem behaviour in a community sample of children aged 5 to 12. Valid sample size for the computed regressions varied from 335 to 339. Parenting practices were assessed using a modified version of the Alabama Parenting Questionnaire (APQ; Frick, 1991). Conduct problems were defined by DSM-IV criteria for Oppositional Defiant Disorder (ODD) and Conduct Disorder (CD), and assessed through the use of the Assessment of Disruptive Symptoms, DSM-IV Version (ADS-IV; Pelham, Gnagy, Greenslade, & Milich, 1992). Both mothers and fathers acted as informants for the two questionnaires. Preliminary analyses suggest that the parenting practices of maternal involvement, poor monitoring and supervision of children's activities, as well as inconsistent discipline, are significantly associated with number of CD and number of ODD symptoms in children. The magnitude of these associations varied according to whether it was the mothers or the fathers who reported the practices. In contrast with previous research in a clinical sample, results did not show a significant child age trend in the associations. This project has significant implications for the parent training component of the psychosocial interventions that are currently used for CD and ODD.

T17. Time Course of Linguistic Processing as indexed by Event-Related Brain Potentials (ERPs)

Randy Lynn Newman and John F. Connolly

While most theories of visual word recognition agree that access to meaning is mediated by phonology, some theories contend that words can access their meaning directly, without phonological mediation. ERPs, unlike behavioral measures of word recognition, provide an on-line measure of cognitive processing without requiring overt behavioral responses. Indeed, a number of ERP components have been found to be sensitive to orthographic, phonological, and semantic information. The goals of the present study were to disentangle orthographic and semantic processing, and to define phonology's role on route to accessing semantic information. ERPs were recorded to the terminal words of visually presented sentences. Phonological/ orthographic and/or semantic characteristics of terminal words were manipulated in 4 conditions in which: (1) the terminal word was the highest cloze probability ending (e.g. Larry writes with his left **hand**.) (congruent); (2) the terminal word was orthographically inappropriate, but was phonologically identical to the highest cloze probability ending (e.g. The game was called off when it started to **rayne**. [rain]) (Pseudofoil); (3) the terminal word was orthographically appropriate, but was semantically incongruent to the sentence context (e.g. The pizza was too hot to **sing**. [eat]) (incongruent); or (4) the terminal word was orthographically inappropriate and was phonologically distinct from the anticipated highest cloze probability ending (e.g. The teacher wrote the problem on the **heet**. [board]) (incongruent-nonword). A negative peak was observed in the 250-300 msec range (the N270) in three conditions in which the orthography was unexpected. That is, even if the orthography was correct, as in the congruent condition, it was orthographically distinct from the highest cloze probability ending for that sentence. A second, left temporo-parietally distributed negative peak in the 350-450 range (the N400), was observed in both congruent conditions. Interestingly, the N400 was absent in the WR condition, suggesting that phonology plays an important role in mediating access to semantic information. Overall, these findings demonstrate the ability of ERPs to measure subtle aspects of linguistic processing.

T18. Transcallosal White Matter Activation Revealed by Functional Magnetic Resonance Imaging During Reading

Ryan C.N. D'Arcy¹, John F. Connolly^{1,2}, Frank P. MacMaster², Robert Vandorpe³, Larry Gates³, Sultan Darvesh⁴, and Matthias Schmidt⁵ 1. Cognitive/Clinical Neuroscience Unit, Department of Psychology, Dalhousie University; 2. Department of Psychiatry, Dalhousie University; 3. Department of Radiology, Dalhousie University; 4. Department of Medicine (Neurology and Geriatric Medicine), Dalhousie University; and 5. Department of Diagnostic Imaging, The Hospital for Sick Children, Toronto

Functional MRI has been used to demonstrate gray matter activation associated with sensory and cognitive functions. We provide the first evidence of fMRI activation in white matter (1.5T). This activation was observed in commissural fiber tracts passing through the splenium of the corpus callosum during sentence reading. The time course data from white matter voxels revealed that, like gray matter activation, the responses were time-locked to the stimulus presentation. However, as expected, the white matter peak amplitudes were markedly attenuated relative to the responses in the visual cortex. Reliability of the white matter activation was established by replicating fully the effect in a participant following a six-month interval. The active white matter regions localized the commissural fibers that connect functionally specialized visual word form areas. These findings provide the first noninvasive identification of the neural architecture involved in sentence reading.

T19. An investigation of the phonological processing in Chinese language with ERPs

Jingtian Wang and John Connolly

Two unique features of Chinese language draw much attention and interest in psycholinguistics. The first is that Chinese written language is pictographic. There is an opaque relationship between the pronunciation and physical appearance of a Chinese character. For example, homophones have the exact same pronunciation but completely different script feature and meaning. This opacity of the script-to-sound relationship has led to a widespread assumption that readers of Chinese can arrive at the mental lexicon by identifying a whole character visually, without any involvement with phonological coding. The issue of whether there is phonological code involved in Chinese reading was investigated with the recording of ERPs. The other specific feature of Chinese language is that it is a tone language. Most Chinese characters consist of monosyllables with an affiliated lexical tone and the tonality of a monosyllable is lexically meaningful. Though the perception of tone plays a major role in Chinese speech comprehension, little is known about its underlying electrophysiological basis. What's more, in behavioral studies, there is a major issue of whether tone perception is of importance in Chinese reading comprehension since there are more errors in tone discrimination tasks during Chinese reading. My future experiments will mainly focus on the above issues with the recording of ERPs, the measurement of brain electrical activity that is time-locked to the underlying linguistic processing.

T20. Magnetic Resonance in Psychiatry

Frank P MacMaster, Shakeela Khan, Vivek Kusumakar

In reference to the invention of the stethoscope, P.A. Piorry remarked "If this method had only a quarter of the utility attributed to it...it would still be one of the most precious discoveries of medicine". The same can be said of magnetic resonance (MR). MR technology offers exciting insight into neurobiology, and the main modalities and their application will be reviewed here. MR is based on the application of a magnetic field and its modulation with a radio-frequency pulse. Magnetic resonance imaging (MRI) offers insight into regional morphology. Magnetic resonance spectroscopy (MRS) offers a non-invasive brain biopsy of neurochemistry. Functional MRI (fMRI) offers a window into brain function via regional brain activation and oxyhemoglobin and deoxyhemoglobin exchange. Diffusion tensor imaging (DTI) tracts white matter coherence and will be especially useful in studies of development. MR, as it uses non-ionizing radiation, is an especially valuable tool for the investigation of pediatric populations. Potential applications in neural development, neuropathology and drug trials will be discussed.

T21. Neurobiological Discriminations of ADHD and Bipolar Disorder

Shakeela Kahn, Frank MacMaster, John Connolly, Normand Carrey, Gail Eskes, Larry Gates, Stan Kutcher, Vivek Kusumakar

Both Attention Deficit Hyperactivity Disorder and Bipolar Disorder are characterized by behavioral and cognitive deficits that can negatively impact on crucial aspects of life such as academic, social and family functioning, as well as enormous financial burden on society. The goal of treatment is to intervene as early as possible and interrupt the course of the illness. Unfortunately, childhood onset of Bipolar disorder often presents with symptoms similar to ADHD (attention deficit hyperactivity disorder). This can result in misinterpretation of attention deficit and impulsivity and misdiagnosis of both conditions in childhood. The consequences of misdiagnosis can be severe. Thus, early, accurate detection of ADHD and bipolar disorder is vital to the management of the illnesses in the long term.

The current management of this problem involves waiting and monitoring the course of the illness; an undesirable option given the evidence for neuronal deterioration and severity that accompanies the natural progression of bipolar disorder (Drevets et al., 1997). One focus of the Dept of Psychiatry at the IWK Grace is to develop neurobiological diagnostic tools to distinguish between ADHD and bipolar disorder at an early age, with the goal of intervening as early as possible. These research directions as well as past research in the area will be discussed.

T22. An evaluation of the Bispectral Index (BIS) as a valid monitor of depth of anaesthesia

Deanna Braaksma, Allen Finley, John Connolly, Michael Houlihan and Patrick McGrath

The BIS index monitor (BIS) (Aspect Medical Systems Inc.) is a relatively new technique that Anaesthesiologists might use to measure anesthetic depth. Aspect claims that BIS is the first clinically proven, direct measure of the effects of anesthesia on the conscious state of patients. However, BIS was not developed to assess cognitive processes. It was merely designed in reference to an observer's assessment of depth of anesthesia (e.g., autonomic responses, movement, blood concentrations). Is the BIS monitor a valid clinical tool for assessing consciousness during anaesthesia? Two approaches are proposed to provide anesthesiologists with stronger, better evidence to base their clinical decisions regarding the BIS monitor: (1) asystematic review/ meta-analysis to comprehensively synthesize the literature on BIS index values and various outcome measures of anesthetic depth, and (2) experimental studies examining sensory and language processing using event-related potentials during anesthesia in adult volunteers at various BIS index values.

T23. 1 bump, 2 bump, old bump, new bump: Using event-related potential maps to identify functionally independent components in ERPs.

Michael Houlihan

Event-related potentials (ERPs) are the electrical signal recorded from a scalp that has been time-locked to a particular event and averaged across many repetitions of the event. ERPs provide a signature of the cortical processing of an event or stimulus with excellent temporal resolution. Advances in computing power and electronics in general have made it possible to record ERPs at many locations over the scalp, introducing an improvement in spatial localization of cognitive processes. ERPs are analyzed by examining the amplitude, spatial distribution and timing of the amplitude. ERP components are usually identified visually as peaks and valleys at each location. If ERP components overlap temporally and spatially they are difficult to impossible to separate. Several solutions to the separation of components have been tested over the years with none adequately filling the role. Recently, an approach to "blind separation" called independent component analysis (ICA), has been applied to neurophysiological signals. ICA has been used with ERPs and fMRI for identifying independent patterns in the data, as onset and duration measures and for the elimination of artifact (muscle activity; EOG; cardiac signals). Examples of the use of ICA will be presented and discussed.

T24. Model of reading aloud: computational and fMRI investigations.

Yannick Marchand

The ability to transform orthographic (letter string) information into phonological (speech sound) information underlies our ability to read words aloud, and it is one of the most important skills we learn. It is often argued that various facts about reading is explained by a model which assumes the existence of a dual-route architecture (lexical and nonlexical routes from print to speech). In this talk, we present some possible cross-fertilizations between a single-route computational model of reading aloud and functional neuroimaging.

T25. Strategy versus ability in speech perception: Behavioural and ERP studies.

Catherine Mayo

Early speech perception research centred on the impressive abilities of the human perceptual system. In particular, such research made much of listeners' tendency to perceive gradually varying speech stimuli in terms of a limited number of meaningful phonological categories. Current ERP research has focussed on determining neural correlates of this ability.

However, more recent perceptual research has shown that while normal listeners show very similar perceptual abilities, the strategies which individual listeners employ to reach a percept will often differ. This talk will discuss current behavioural studies, which have shown developmental differences in perceptual strategies. The talk will conclude with a discussion of the potential for ERP studies to further our understanding of perceptual strategy differences.

T26. Does Vagus Nerve Stimulation for the Treatment of Medically Refractory Seizures Enhance Memory?

Marsha Penner², Ivan Valdivia², Jeannette McGlone^{2,3}, Mark Sadler¹ and Dave Clarke¹, Department of Pharmacology¹ and Departments of Psychology², Dalhousie University & QEII HSC³.

Epilepsy can have a profound impact on all aspects of a person's life. A novel treatment, vagus nerve stimulation (VNS), has recently been introduced to treat medically refractory seizures. Despite some promising effects on seizure frequency, little information is available regarding the effect of VNS on cognitive functioning. A recent report in Nature Neuroscience indicates that VNS enhances recognition memory in human subjects (Clark et al, 1999). However, this report has not been substantiated by other groups. The present study examines memory function in a group of patients being treated with the vagal nerve stimulator for medically refractory seizures. Subjective and objective memory measures, including the Memory Observation Questionnaire and the Weschsler Memory Scale-III are administered immediately before implantation of the stimulator and again at a follow-up visit one year later. Outcome measures from the VNS patient group are compared to two control groups, including a group of patients who have undergone temporal resectioning, and a group of patients maintained on antiepileptic medication.

Clark et al 1999 Nature Neuroscience
Where does the vagal nerve go in the brain to stimulate memory?

T27. Getting Hip on the Chip

CS7 mouse
Elliott Marchant, Donna Goguen, Marleen deGroot, and Ben Rusak.

This talk will outline new technology that provides efficient access to genetic information using miniaturized high-density arrays of oligonucleotide probes. This technology is currently being used by the Rusak lab in collaboration with Novartis Institute for Functional Genomics and Affymetrix, to explore the molecular impact of light on the mouse suprachiasmatic nucleus.

T28. Meals on Wheels

in PWT state
Marleen de Groot, Elliott Marchant and Ben Rusak

orexins
Animals exposed to schedules of temporally restricted food access will begin to anticipate the arrival of food after a number of days on the schedule, typically by increasing activity, such as wheel-running. This anticipatory activity is regulated by a self-sustaining, endogenous pacemaker that is both structurally and functionally distinct from the master circadian pacemaker. The identity of this food-entrainable pacemaker and the nature of the afferent signals that affect it remain unknown. We are investigating which brain regions may be involved in the acquisition and expression of food anticipatory activity. We have recently assessed patterns of immediate-early gene (IEG) expression in various brain regions in mice who have been acutely food deprived or exposed to a repeated daily schedule of food access. The preliminary data showing the effects of refeeding on IEG expression in these two groups of mice will be discussed.

T29. Intentional relations and divergent perspectives in social understanding

Sakurai et al 1999 Regulatory Peptides 85: 25-30.
John Barresi

People are very skilled at applying folk-psychological concepts uniformly to self and other. Thus, I understand your being in love in much the same way that I understand my being in love. But the information I have about you being in love is hardly the same as the information that I have about me being in love. This informational difference can sometimes lead to divergent interpretations of human actions between actors and observers. Not surprisingly, social psychologists, always on the lookout for differences rather than commonalities, have stressed this divergence rather than the commonality in social understanding, naming it the 'actor-observer effect'. In the present paper, I extend a model of social understanding developed by Barresi & Moore (1996) to account for divergent interpretations of actions, not only between actors and observers, but also between members of different social groups.

T30. Lateral Hypothalamic Lesions Disrupt Time-Place Learning in Rats

C. Christensen, A. Gregus, K. Luedeman, S. MacKinnon, B. Oliver, J. Pitman, J. Taggart, Marleen de Groot & Elliott Marchant

Time-place learning refers to situations in which an animal learns to associate a particular time of day with a particular location. The acquisition of the association between temporal and spatial information has previously been studied in various species. In order to investigate whether the lateral hypothalamus (LH) plays a role in time-place learning, intact ($n=6$) and LH-lesioned ($n=4$) rats were trained to obtain food at two distinct locations, each at a different time of day. A standard two lever T-maze was used in this study. Rats were rewarded for lever presses on the left side during morning session and on the right side during afternoon sessions. All 10 rats learned the task, however, omission tests showed that LH lesioned animals performed worse than intact animals, suggesting that lesioned animals were using a different strategy. The results indicate that the LH may be an important structure involved in time-place learning.

T31. Is there a relation between Apolipoprotein E gene status and neuropsychological memory test performance? An analysis of the Canadian Study of Health and Aging

Jennifer Klages and John D. Fisk

The current identified risk factors for Alzheimer's disease (AD) have insufficient predictive value. One is the presence of E4 allele of the Apolipoprotein (APOE) gene. Poor performance on neuropsychological tests is another. We examined the relation between these two risk factors in a population-based sample of people over age 64. Participants were enrolled in the Canadian Study of Health and Aging (CSHA), a nationwide study of the prevalence of dementia that involved over 10,000 people. APOE status was determined in 1578 of which 1188 had no E4 allele and 390 had at least one E4 allele. From this group, 1123 had no cognitive impairment (NCI), 268 had cognitive impairment but not dementia (CIND) and 187 had dementia. Neuropsychological test performance of subjects with and without the E4 allele was compared within these diagnostic categories. NCI subjects who had an E4 allele performed worse on two memory tests: the Buschke Cued Recall Test ($p<.004$) and the Rey Auditory Verbal Learning Test ($p<0.05$). APOE status was not related to test performance for the CIND or demented groups and was not related to performance on any other tests. APOE status and memory test performance have some relationship in nondemented elderly. Together, these factors may help to predict the development of dementia at followup.

What tasks are in the Buschke test? Rey test.
Buschke test?

T32. Alcohol effects on Risk-Taking during Video Lottery Terminal (VLT) Play Among Pathological and Non-Pathological Gamblers

Mike Ellery, Sherry H. Stewart, Pamela Loba, James R. Blackburn, & Raymond Klein

As part of an ongoing study, four risk-taking variables were examined in order to test the hypothesis that alcohol consumption would lead to increased risk taking during gambling play, particularly for pathological gamblers. Participants were regular video lottery terminal (VLT) players, recruited by newspaper advertisement in the community. An equal number of the participants run thus far were assigned to alcohol and no alcohol control conditions (N = 10 per beverage condition). Eight of the 20 (N = 4 per beverage condition) met criteria for probable pathological gambling, as assessed by the South Oaks Gambling Screen (SOGS; Lesieur & Blume, 1987), and the remaining 12 scored below the SOGS cutoff, indicating non-pathological gambling status (N = 6 per beverage condition). Participants were randomly assigned to either the moderately intoxicating dose of alcohol condition, or the control beverage condition (mix only). Following beverage consumption and absorption, participants were allowed to play a poker game on a commercially available VLT for up to 30 minutes. Four indices of risk-taking were recorded throughout gambling play: (a) number of times participants hit the "powerbet" button (which allows gamblers to double their bet size after having viewed only two cards of their five card poker hand); (b) total money inserted into machine during entire 30 minutes of play (in Canadian dollars); (c) average bet size during the last 5 minutes of play (also in Canadian dollars); and (d) total time out of a possible 30 minute play period that subjects chose to play machines (measured in minutes). Results of preliminary planned comparison analyses provide evidence that: (1) alcohol consumption increases risk taking behaviour while gambling, but only among pathological (and not non-pathological) gamblers, and (2) increased risk taking is displayed by pathological relative to non-pathological gamblers during VLT play, but only when they are intoxicated, and not when they are sober. Theoretical and clinical implications of the findings will be discussed.

T33. Trade-offs between current and future reproduction in the cricket, *Gryllus integer*

Kelly Shoemaker and Shelley Adamo

In order to maximize lifetime reproductive success, organisms must often make trade-offs between current and future reproductive effort. We examined these trade-offs in the cricket, *Gryllus integer*, in response to predation risk and infection: two factors that could potentially reduce future reproductive output to zero. We found that *G. integer* females laid more eggs when the egg-laying substrate was covered than when exposed. However, *G. integer* females did not lay more eggs following infection with a potentially lethal pathogen, *Serratia marcescens*. These results suggest the *G. integer* females regulate oviposition based on predation risk, but that it is not influenced by immune status.

T34. Accumulation, Reactivity, and Adaptation Models of Event and Mood Relationships

Robin L. Patterson & Darcy A. Santor

Many studies examining the relation between events and mood employ a 2-wave panel design in which events and mood are assessed at two time points. This assumes that the effects of events accumulate and persist until mood is assessed at follow-up. This design is inadequate to assess mood reactivity or individual adaptation. The present study examined three models of event and mood relationships: accumulation, reactivity, and adaptability. Participants (N=102) completed mood and event checklists twice weekly for 10 weeks. These checklists involved recording events and tracking moods and behaviors. The results showed little evidence for the accumulation model. Results supported models that tested reactivity and adaptation. Both positive and negative events influenced negative moods; however, few negative events resulted in lasting mood disturbances. Mood disturbances that were precipitated by a negative event lasted less than six days. Results have implications for how events are assessed and the types of proposed models.

T35. Effects of Chronic Methylphenidate Administration at Several Stages of Development in CD-1 Mice

Melanie McFadyen¹, Marsha Penner², Stephen Lewis¹, Normand Carrey³, & Richard Brown¹, Departments of Psychology¹, Pharmacology², and Psychiatry³, Dalhousie University, Halifax, NS, Canada.

Although methylphenidate hydrochloride (MPH, Ritalin) has been used in the treatment of attention deficit hyperactivity disorder (ADHD) for several decades, research examining the potential effects of MPH on neurobehavioural development is lacking. We have conducted several experiments that begin to systematically detail the effects of different doses of MPH administered at different times during development (prenatal, neonatal/infancy, and prepubertal). Mice exposed to MPH (5 mg/kg) prenatally (embryonic days (E) 8-10 or E12-14) exhibited significant increases in spontaneous locomotion when tested at postnatal days (PND) 3-11. No differences were observed in tests of neuromotor reflexes such as righting, negative geotaxis, or forelimb grip strength. In adulthood, the mice exposed to MPH from E8-10 exhibited significantly increased exploration in both the open field and the elevated plus maze, as well as decreased anxiety in the elevated plus maze. Chronic MPH administration (5 or 20 mg/kg) during postnatal development (PND3-11) resulted in significant increases in locomotion, but had no effect on ultrasonic vocalizations when pups were tested one hour after each injection. In adulthood, these mice performed equally well in the Morris water maze, showing no deficits in either learning or memory in this cognitive spatial task. No doses of MPH (2.5, 5, 10, 20, 40, or 80 mg/kg) administered during prepubertal development (PND26-32) had an effect on spontaneous locomotion, exploration, fear or anxiety in the open field and elevated plus maze when the mice were tested at PND 33. There was no effect of any dose of MPH on learning and memory in the Morris water maze when mice were tested from 34-37 days of age. The results of our experiments suggest that while acute administration of MPH does elevate locomotor activity, chronic administration has no enduring deleterious effects on neuromotor reflexes, learning or memory. Experiments examining longer administration times, MPH exposure at different stages of development, and effects of MPH in animal models of ADHD are necessary in order to fully establish the safety of MPH exposure during development.

T36. Effects of the Benzodiazepine Receptor Agonist Chlordiazepoxide & the 5-HT_{1A} Antagonist WAY100635 on Behaviour in the Rat Elevated Zero Maze are not Influenced by the Light/Dark Cycle.

Lianne Stanford, Scott M. Weiss* and Kelly HJ. Stanhope* *Department of Neuropharmacology, Cerebrus Ltd., UK

The rat elevated zero maze is a validated model of anxiety (Shepherd et al. 1994 Psychopharm 116:56-64). In this test, rats administered anxiolytic drugs typically show a behavioural profile of increased exploratory activity (line crosses & head dips), increased time in the open areas, and less risk assessment (stretch-attends). In mice, 5-HT_{1A} receptor antagonists have altered profiles on the elevated plus-maze when tested during different portions of the light/dark cycle (Rodgers et al., 1998 J Psychopharm 12:A32). To investigate this further, rats housed under normal or reversed light/dark cycles were tested with chlordiazepoxide (CDP), a benzodiazepine receptor agonist or with WAY100635, a 5-HT_{1A} receptor antagonist. CDP (4 mg/kg) treated rats showed a similar anxiolytic profile in both housing groups. WAY100635 (0.01 to 0.3 mg/kg, sc: 30min) was without effect on rats from either housing group. These data show that the time of testing relative to the diurnal cycle does not alter the profiles of either CDP or WAY100635, and question the assumption that time of testing is a critical determinant of the anxiolytic properties of 5-HT_{1A} receptor antagonists.

T37. The identification of pheromonal components in male mouse urine which initiate puberty acceleration in female mice

Poster Heather Schellinck, Allison Clarke, Richard Brown and Michael Wilkinson

The pheromonal components of male mouse urine which accelerate puberty in female mice have not been clearly identified. To clarify this issue, we examined the effect of a number of compounds on puberty acceleration and neural activation of the accessory olfactory bulb (AOB). During postnatal days 21-28, juvenile female mice, were exposed to male urine, major urinary proteins (MUPs), MUPs+ligands attached (2-sec-butyl-dihydrothiazole:SBT and 2,3-dehydro-exo-brevicommin:DHB), SBT/DHB alone or saline. Puberty acceleration occurred in mice exposed to male urine or MUPS+ligands but not in mice exposed to SBT/DHB alone or MUPS alone or saline. Immediate early gene expression (c-fos) was significantly higher in the AOB of mice exposed to male urine or MUPS+ligands than in those mice exposed to saline, SBT/DHB or MUPS alone. These results indicate that exposure to specific components of male mouse urine not only induces puberty acceleration but also activates the neural pathway which initiates such a effect.

T38. Evolutionary origins of morality: Is there an ancestral 'logic' to the modern mind?

William Michael Brown

Small sympathy & altruism

Selfish gene theory has difficulty explaining the existence of moral impulses. Evidence from evolutionary economics suggests that individuals can forgo the temptation to cheat due to the emotions of guilt and sympathy. However genes predisposing the development of these prosocial emotions would be at a fitness disadvantage in a population of exploiters. How did sympathy evolve if it is costly to the altruist? One possible pathway is the formation of cooperator-cooperator alliances. If group formations were based upon intrinsically unfalsifiable sympathy signals, non-kin altruism may be evolutionarily stable. Perhaps the benefits to the genes involved were accrued at the within- and between-group levels of selection. Preliminary evidence that humans detect an individual's likelihood to cooperate will be presented.

T39. Remote computing: Internet access from Far Away Places

Ron Hoffman and Richard Brown

Reliable access to the internet has become a requirement for today's academics and we take it for granted in our daily work at the university. However, when you take your work (and office) on the road (or the high seas), you need to make special arrangements to maintain that internet access. Technological advances of the recent years hold much promise for remote workers. Wireless communication is particularly enticing. In an entertaining illustrated presentation, we will discuss what worked reliably, what did not, based on our recent travels.

T40. We do PowerPoint, we do web pages, now what? Digital Video!

Ron Hoffman, Vivien Hannon (ACS), John MacDonald (ACS), David Patriquin (Biology)

Recent technological advances in what has become known as Information Technology have changed the way we work. Chalk and blackboards are rapidly being supplanted by PowerPoint presentations, and we expect students to use that technology, too. A primary source for information is now the web (although too many students seem to see it as the only source), and our graduates need to know how to create their own web sites when they hit the job market. Now, new technologies are bringing digital video to the desktop, and we are just beginning to see the impact. We discuss these technologies, show some examples of what can be done, and ask if we should add another skill to our IT skill set.

T41. Mental Geography Meets Mental Chronometry in an Exploration of the Law of "Prior Entry"

Ray Klein, David Shore, J. Gati & R. Menon

Abstract: It has been suggested that psychology was borne from the union of interests in mind (philosophy) and brain (physiology). Here we seek to unite two methodologies with strong connections to mind (mental chronometry) and brain (neuroimaging) in the expectation that together they will be able to do more than either could alone. Our study concerns Titchener's Fourth Law of Attention: "The stimulus, for which we are predisposed, requires less time than a like stimulus, for which we are unprepared, to produce its full conscious effect" (1908, p251). We will illustrate how this law of "prior entry" can be explored and precisely measured psychophysically by obtaining temporal order judgements for pairs of stimuli, one of which is attended. We will then present preliminary fMRI data suggesting that specific brain regions are involved in performing these judgements. Finally we will propose how event-related fMRI can be used to test Titchener's "Law of Prior Entry" and how such neuroimaging studies of prior entry can be useful in demonstrating the potential of latency-resolved fMRI.

T42. There is More to Tactile Perception than Simply Touch

David I. Shore*, Emily Spry^, Charles Spence^ * Rotman Research Institute; Baycrest Centre for Geriatric Care; University of Toronto ^ Department of Experimental Psychology; Oxford University

We explored the role of body position in the perception of simple tactile stimuli. The observer was presented with brief tactile vibrations to the left and right forefinger and was required to indicate which hand was stimulated first (using a foot pedal response). The interval and order between the two stimuli was varied randomly in order to produce a psychophysical function from which the Just Noticeable Difference (JND) was derived. This measure indicates how much time is needed in order to indicate reliably which stimulus was presented first. In three separate experiments observers adopted different postures: Hands crossed, hands far apart, and one hand displayed in a mirror. The JND was much larger when the hands were crossed compared to uncrossed and, when hands were close together rather than far apart. The mirror manipulation had no effect on JND. The finding that changing hand position resulted in a change in JND is surprising given that the same receptor surface is stimulated in both conditions. The role of body centred representations in tactile perception will be discussed.

T43. The Rising Cost of OOTLUF

Joe MacInnes, and John McCabe

Recent studies have looked at the trade-off involved in auto-pilot systems. Although automation does reduce the workload of the pilot in important tasks, it also seems to reduce the situational awareness, and produce Out of the loop unfamiliarity, (OOTLUF). This present study tested this trade-off by asking participants to navigate a set of virtual Hebb/Williams mazes, under both manual and automated conditions. Automation was simulated on alternate mazes by having participants learn a route through which they were passively transported using a previous participant's active exploration. The cost of situational awareness was measured as the increase in time to complete a maze based on the method of learning, while workload was measured through simple reaction time to random and frequent probes in both the manual and automatic conditions. Although the OOTLUF, cost of automation was observed in this experiment, the benefits to workload were not as clear cut.

T44. The Negative Consequences of Being a Good Boy Scout

Tracy Taylor-Helmick & Jason Ivanoff

The boy scout motto - "always be prepared" - may lead to rapid responses when a wilderness creature attacks suddenly. But when it comes to reacting to visual targets, responses may be slowed by prior (motor) preparation. We presented participants with two peripheral visual targets to which speeded spatially-compatible button-press responses were required. On a random half of the trials, a Stop Signal was presented at a variable delay following the First Target. Participants were instructed to withhold a response to the First Target when the Stop Signal appeared. Their ability to do so varied with the interval between the First Target and the Stop Signal appearance: The longer the delay, the less likely the participant was to withhold the prepared response. We examined how reaction times to the Second Target varied with the probability of canceling a response to the First Target. Although the results are preliminary, they suggest that preparing a response to the First Target slows the speed of making that same response to the Second Target. More importantly, the degree of slowing is related to the degree of prior motor preparation: The more prepared the First Target response is (i.e., the more difficult it is to cancel), the greater the inhibitory aftereffects may be.

T45. What 'coon you say about brain circuits using local inactivation?

Susan Boehnke and Doug Rasmusson

The raccoon has proved a great model for investigating cortical reorganization after peripheral denervations (e.g. amputation) due to their enlarged representations of the forepaw digits. In the raccoon cortex, a large sensory evoked potential (SEP) occurs in response to electrical stimulation of the corresponding digit - an "on-focus" response. Stimulation of a neighbouring digit evokes a smaller SEP - an "off-focus" response. It has been hypothesized that the off-focus response is conveyed through intracortical circuits from one digit representation to another and that these corticocortical pathways may contribute to the patterns of cortical reorganization observed in this species. We examined this hypothesis by recording on- and off-focus SEPs from 2 neighbouring digit cortical regions and examining the change in SEPs during a selective, reversible inactivation of one region using lidocaine and tetrodotoxin. Although the drugs blocked the "on-focus" response in the inactivated area, this did not result in a significant decrease in the amplitude of the "off-focus" response from the neighbouring cortical digit area. This indicates that changes in corticocortical connections may not play a major role in reorganization following amputation.

T46. Quality of Life in Patients with Vagal Nerve Stimulators

Ivan Valdivia, Marsha Penner, Jeannette McGlone, Shirley Munk, David Clarke, and Mark Sadler

People with chronic epilepsy may experience a quality of life that is not optimal. Patients whose seizures are resistant to medication and who are not eligible for brain surgery may now be able to use Vagal Nerve Stimulation (VNS) as an alternative treatment. While the study of Quality of Life in persons with epilepsy is researched, information regarding the Quality of Life of patients using VNS is not. This study aims to examine Quality of Life in a VNS implantation group and two control groups: 1-those who have undergone temporal lobe resectioning, and 2-those who are maintained on medications alone. Two separate measures of Quality of Life (i.e., the Quality of Life in Epilepsy-89, and the Quality of Life Questionnaire) are administered immediately before implantation of the VNS and again at a year follow-up. Preliminary data will be presented.

T47. The evolution, history, and development of consciousness: A sketch

John Barresi

In the present talk I will give an overview of the evolution, history and development of consciousness. Obviously, it will only be a sketch. Basically, the issue is 'why consciousness'? What's consciousness for? How did it evolve to the level we find it in the human species? How did consciousness change historically in the human species? And, how does consciousness develop during the life of an individual human person? I promise to give answers to all of these questions. Whether you will believe my answers is quite another issue. But what can you expect from a ten-minute sketch?

Final lecture & history memory

T48. Neuropharmacology and neurobiology of general anesthetics: the mystery of human consciousness

Deanna Braaksma

Cholinergic system

Can the nature of consciousness be quantified, located, and understood within the human brain? Prevention of conscious awareness is a main goal of general anaesthesia. Examining (1) the molecular mechanisms of action of diverse anesthetic agents on various sites of the brain, (2) the similarities and differences between neuronal systems of anaesthetized and non-anaesthetized brains, and (3) the effects of anesthetics on higher brain functioning using functional brain imaging may be a fundamental empirical first step toward the eventual understanding human consciousness. The current presentation will outline what is known about the neurobiology and pharmacology of general anesthesia (e.g., NMDA receptors, Ach neurotransmitters, neuronal proteins, cortical-thalamic interactions, whole-brain metabolism), and will attempt to relate it to current models of consciousness (e.g., Rosenthal's metarepresentation, 1990; Tulving's multiple systems, 1998).

T49. "Now you see it, now you don't: Or did you?"

John Christie and John Barresi

Part of a recent experiment by John Christie and John Barresi will be presented as an example of the use of psychophysical methods in studying consciousness. The pitfalls, and controversies around these methods will be mentioned and hopefully incite much painful thought. Questions will be entertained but not necessarily answered.

T50. The Hard Problem of Consciousness: Bridging the Explanatory Gap

Bill Matheson

Many (perhaps most) psychologists and neuroscientists believe that an adequate, materialistic understanding of brain and behaviour will entail an understanding of the mind and consciousness. There will be no irreducible or incompatible residue left to explain once our empirical investigations, and their theoretical interpretation, are complete. Some (perhaps most) philosophers, however, are skeptical of this optimistic outlook. They believe that the mind, and particularly consciousness, have properties which cannot be brought within the scope of scientific explanation. There is an explanatory gap between a complete physical story, and one which includes such conscious phenomena as qualia and values. Although we can expect that the easy problems of perception, learning, discrimination, and so on, will be solved through scientific methods, the hard problem of consciousness will prove to be much more difficult, and perhaps impossible.

P1. An examination of strain differences in nest building in virgin, pregnant and lactating C57BL/6 and DBA/2 mice (*Mus Musculus*).

Tamara L.Y. Bond¹, Richard E. Brown¹, Christianne Hawken¹ and Paul E. Neumann².
Departments of Psychology¹ and Anatomy & Neurobiology², Dalhousie University.

C57BL/6 (B6) and DBA/2 (D2) mice differ in maternal behaviour and nest building. However, the results on nest building have been contradictory. Although lactating B6 females spend more time nest building than lactating D2 females (Brown et al., 1999, *Physiol Behav*, 67:599-605), pregnant D2 females build better pre-parturient nests (Broida & Svare, 1982, *Physiol Behav*, 25:153-157). To resolve this discrepancy, we recorded nest quality in virgin, nulliparous, pregnant and lactating females. Virgin B6 and D2 females were housed with either a male or female of their own strain for fifteen days. Females were then housed singly and their nests rated daily on a scale of one to four. Some of the females of each strain that were housed with males failed to give birth and some litters died before the pups reached four days of age. There were no strain differences in the nest ratings of virgin females or females who did not give birth, nor did these groups differ. Therefore, these two groups were collapsed into one group of nulliparous females. No differences were observed between strains in the proportion of pups surviving per litter or in pup weight gain. B6 and D2 females whose litters survived built better nests than nulliparous females. Among females whose litters survived, B6 females built better nests than D2 females both before and after birth. Although both strains increased their nest building on the day of parturition, there was a strain by day interaction that resulted from a dramatic increase in nest building by D2 mice on the day of parturition. However, the increase in nest building by D2 mice at parturition was not large enough to ameliorate the difference in nest building between the two strains. Mothers of both strains whose litters died built better nests than nulliparous females but nest ratings did not differ between strains. The nest ratings of D2 mothers whose litters died did not differ from those of D2 mothers whose litters survived but B6 mothers whose litters died built poorer nests than B6 mothers whose litters survived. One hypothesis to account for these results is that females of these two strains differ in hormone levels during pregnancy.

P2. The Role of the Frontal Lobes in Implicit Memory

Gail Eskes, C. Szostak and D. T. Stuss

While the role of the frontal lobes in explicit memory is well supported, the findings for implicit tasks are less conclusive. We investigated the role of the frontal lobes in perceptual and conceptual implicit priming. Three memory tasks were given under both implicit and explicit retrieval instructions using word fragment completion, picture fragment completion and category exemplar generation. Four groups were studied: Left dorsolateral lesions, right dorsolateral lesions, medial lesions (left, right or bilateral), normal controls. Word fragment completion priming was impaired by left dorsolateral lesions, with other priming tests unaffected by any lesion. Explicit performance showed a different pattern, with category exemplar cued recall impaired by all lesions. These findings support the role of the frontal lobes in both implicit and explicit retrieval mechanisms.

P3. Criminal Career Profiles as a Function of Psychopathy and Sexual Violence

Angela R. Birt, Stephen Porter, & Mike Woodworth

The problem of sexual violence has become increasingly apparent in recent years. In this study, the complete criminal and conditional release performance histories of 217 federally incarcerated sex offenders and 101 non-sex offenders were examined. Offenders were categorized according to the nature of their sexual violence and all offenders were rated on level of psychopathy (PCL-R, Hare, 1991). Results indicated that (a) certain sex offender groups had higher levels of psychopathy than others, (b) psychopathy exacerbated the violent and non-violent crime committed by all offender groups, and (c) Mixed Rapists/ Molesters were the most dangerous as indexed by their psychopathy prevalence, criminal profiles, and conditional release performance. These findings have implications for the management and treatment of sexual violence.

P4. Effects of parental experience and the presence of the father on pup growth and development in Peromyscus californicus.

Mauro L. Vieira (Universidade Federal de Santa Catarina, Brazil) & Richard E. Brown (Dalhousie University, Canada).

This experiment investigated the effects of the presence of the father and parental experience on pup development and behavior in California mice. Twenty-seven litters of California mice were used as subjects. Litters housed with single experienced mothers (n=9), experienced fathers and mothers (n=9) and inexperienced fathers and mothers (n=9) were studied. Behaviors of pups were recorded every second day from 10 to 30 days of age. The results indicated that the presence of the father did not influence pup behavioral development and physical growth. Pups of first time parents were more active and had advanced locomotor activity in comparison with pups from experienced parents. It is suggested that the parity of the parents has greater influence on the development and behavior in California mice than the presence of father. Financial support: CAPES (Brazil) and NSERC (Canada).

P5. Coding for orientation in the representation of faces

Mary-Ellen Large and Patricia McMullen

A common debate in face and object recognition literature is whether faces and objects share similar processing and representational systems. Experiment 1 investigated whether patterns of response to faces rotated through multiple orientations were similar to those observed for objects. Participants named digital images of faces presented at multiple orientations over a series of 8 blocks. The findings suggested same effects of orientation on rotated faces and object naming during the first block of presentations. However, unlike objects, effects of orientation required many more blocks to diminish. Experiment 2 investigated the kind of representations underlying these practice effects. Participants repeatedly named faces at two orientations. After training, response times were recorded for faces rotated at the trained orientations and two "surprise" orientations. If, like objects, practice effects transferred to faces presented at "surprise" orientations this would suggest that orientation-invariant representations were encoded. Conversely, no transfer of practice effects would suggest that orientation-specific representations were encoded.

P6. An Investigation of Factors Influencing Accuracy at Detecting Deception

Jennifer A. Stapleton & Stephen Porter

Although empirical research has demonstrated that there are reliable verbal and nonverbal cues to deception (e.g., Porter & Yuille, 1996), the ability to detect deceit eludes most people. The purpose of the present study was to investigate possible factors influencing accuracy at detecting deception, including cue usage, personality characteristics, gender, handedness, and confidence. Participants were asked to judge the honesty of eight videotaped speakers relating an account of a personal experience. Half of these tapes were presented audio-visually while the other half were presented with audio only. Participants provided both a baseline report of cues they generally used in detecting deceit and cues they actually relied upon in formulating each judgment. Major findings were that: both target attractiveness and female gender were negatively associated with accuracy; lies were easier to identify than truths; left-handers were better at detecting deceit than right-handers; and the number of overall cues used was positively correlated with accuracy.

P7. Subscale scores on the Child Anxiety Sensitivity Index (CASI) predict unique variance in DSM-IV anxiety symptoms beyond trait anxiety

Elizabeth McLaughlin, Sherry Stewart & Steve Taylor

The CASI is a self-report questionnaire, which surveys children's fear of the physical, social, and psychological consequences of anxiety sensations. Groups of items pertaining to each of these consequences have been empirically shown to form three distinct lower-order factors on the adult version of the scale. Our first goal was to validate this factor structure in the CASI. Secondly, we wanted to determine whether scores on each CASI factor predicted symptoms of anxiety disorder subtypes, after taking out variance due to trait anxiety. The CASI, a measure of DSM-IV anxiety symptoms (Spence Children's Anxiety Scale), and the STAIC were administered to 349 school-aged (7-15 years) children. Scores on each of the three CASI factors predicted unique variance in conceptually related anxiety symptoms.

P8. Differences in Children's Conceptual Strategies When Thinking About Animate and Inanimate Entities

Nicole Blanchet, Philip Dunham, & Frances Dunham

In two experiments, three-year-old children were asked to match standard pictures (e.g., a dog), with either a taxonomic item (e.g., another dog) or a thematic item (e.g., a bone). Animate and inanimate standard pictures were compared on their potential to elicit thematic matching responses. Results from both experiments indicated that, while taxonomic choices predominated overall, the relative frequency of thematic selection was highest in the animate standard condition. These data support recent suggestions that thematic thinking may presuppose taxonomic thinking, and that the progression from taxonomic to thematic thought proceeds more rapidly in some domains of knowledge than others.

P9. One or Multiple Mechanisms Underlying Orientation-Invariant Identification of Objects?

Karen Jaskie and Patricia McMullen

What are the mechanisms underlying orientation-invariant identification of objects? We are investigating the conditions under which orientation-invariant identification of objects occurs, and whether one or several mechanisms underlie that process. Orientation-invariant recognition occurs at the superordinate and basic levels of identity, as well as after practice with naming rotated objects. The underlying mechanism(s) might be based upon the extraction of orientation-invariant features, or, upon the global attributes (requiring intact spatial arrangement of the features) of objects. We presented four blocks of feature-scrambled objects followed by a block of intact, rotated objects, in a verification task where level of identity was controlled. If the mechanism is feature-based, superordinate- and basic-level identification of scrambled objects should be faster than subordinate-level. Furthermore, practice at identifying scrambled objects should diminish orientation effects in the final block. A global-based mechanism would not produce these effects.

P10. Executive Function Abilities and Prudence and Altruism in Preschoolers

Shana Nichols and Chris Moore

The aim of the current study was to assess the cognitive basis for emerging social behaviour during the preschool period. A comprehensive battery of executive function tasks measuring inhibition, attentional flexibility, and working memory, and a future-oriented prudence and altruism task was administered to 58 preschoolers aged 3;6. Age and language partialled correlations revealed that children who were unable to inhibit copying a puppet's actions, and those who could not refrain from peeking during the wrapping of a present were more likely to choose two stickers for later on a delay of gratification for self task. Children who waited longer before peeking were also more likely to choose one sticker over two for themselves in the present. There was no relationship between performance on any EF task and future-oriented sharing behaviour. Contrary to the predicted outcome, there was a significant negative relationship between performance on both inhibition tasks and children's choice to forego an immediate reward. A hypothesis is put forward regarding the possibility that poor inhibitors were choosing two stickers for different reasons than would be expected. Further longitudinal relationships between the EF variables and performance on the future-oriented prudence and altruism task will be discussed as children were also tested at age 4;0.

P11. Effects of Chronic Ritalin Administration on Postnatal Mouse Pup Development

Marsha Penner², Melanie McFadyen¹, Normand Carrey² & Richard Brown¹. Departments of Psychology¹ and Psychiatry², Dalhousie University.

Methylphenidate hydrochloride (MPH, Ritalin) is a psychostimulant that is used primarily in children for the treatment of ADHD. Despite having been used for several decades, the consequences of chronic MPH administration has not been well characterized in the developing nervous system. Thus, our experiment investigated the effects of repeated administration of MPH on development using several tests from the Dalhousie Developmental Test Battery including isolation-induced ultrasonic vocalizations (UVs), spontaneous locomotor activity, and several tests of neuromotor development in outbred CD-1 mouse pups. From postnatal days (PNDs) 3 to 11, mice received daily injections (s.c.) of, 1) a high dose of MPH (20mg/kg), 2) a low dose of MPH (5mg/kg), 3) an equivalent amount of saline, or 4) no injection. Mice were tested 30-70 minutes post-injection on PNDs 3, 5, 7, 9, and 11. Both the high and low doses of MPH resulted in significant increases in locomotor activity at all ages tested, but had no significant effect on the number or duration of UVs emitted. Contrary to our expectations, performance on tests of neuromotor development was not clearly affected by either dose of MPH. Furthermore, MPH treatment had no effect on body weight. These results suggest that chronic MPH treatment has no short-term effects on the developing nervous system.

P12. Effect of diazepam on the behaviour of "anxious" C57BL/6 mice in the step-down passive avoidance task

Sean McCabe, Jana Podhorna, and Richard Brown

In humans, anxiety impairs learning and memory in a variety of tasks. C57BL/6 mice display higher levels of anxiety than other strains of mice in various models of anxiety such as the elevated plus maze and open field. Moreover, C57BL/6 mice show memory impairments in models such as the passive avoidance task that involve a fearful or "emotional" stimulus. Diazepam has been used to treat anxiety in humans, but has been found to possess amnesic properties by impairing the acquisition process. In animals, benzodiazepines administered before learning trials in passive avoidance tasks impair memory retention but not acquisition. The present study examined the effects of diazepam on acquisition and retention of the passive avoidance task in male and female C57BL/6 mice. Diazepam or saline was administered to the mice 30 min before the learning trials. Female mice appeared to be more sensitive to the sedative effects of diazepam than males. However, diazepam did not impair acquisition or retention of the passive avoidance task in either sex. These results suggest that diazepam does not produce the amnesic affect in anxious C57BL/6 mice that is seen in other strains of mice.

P13. Event Resolution Imaging: Overcoming the limitations of signal averaging in ERP research

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The interpretation of ERPs has typically relied on signal averaging to extract information from background noise (i.e. EEG). A new signal detection method, event-resolution imaging (ERI), has demonstrated the ability to extract information from the EEG with little or no averaging. ERI uses a three-step process of pattern learning, pattern recognition, and validation. During pattern learning, multiple epochs from two distinct conditions are analyzed to detect and subsequently discriminate robust patterns in space (multiple electrode sites), time, frequency, and phase. This information is encoded as a template that is subsequently used to analyze unclassified, unaveraged or few-trial averaged EEG data. We tested the feasibility of ERI by reanalyzing data recorded during performance in a neuropsychological reading test (PALPA; D'Arcy, Connolly, Eskes, 1999). Using traditional averaging, correct and incorrect trials in the PALPA were differentiated on the basis of a P300 response to correct trials. For each epoch analyzed, ERI generated a high signal-to-noise waveform and subsequent classification. Initial results on 5 trial averages indicated an average accuracy level of 72.3%, and results for a single trial analysis indicated an average accuracy of 62.3%. Data of a learning disabled participant, who exhibited no P300 activity in averaged data, were classified with 76% accuracy. Interestingly, for this individual ERI found that 100% of correct trials failed to exhibit correct-trial template features (i.e. no P300) as seen in all healthy individuals. ERI appears to be a powerful tool for use in basic and clinical research.

